Fluency bank and scheduling

Two or three questions in 5 to 8 minutes, correction included. The first markers indicate when the prerequisites are available; the table offers spaced reviews, to be adjusted according to errors. The routine does not replace problems or time spent working on them.

Questions grouped by first use

R01 · Calculation and subscripts. Première sequences; available from S2. For $u_n = 3n - 2$, calculate u_4 then u_{4+1}.

R02 · Mental calculation. Powers; available from S2. Calculate 2^{-3} & $\frac{2^5}{2^3}$ without decimals.

R03 · Context. Proportions; available from S2. A quantity falls by 20 %, then increases by 20%. Does it return to its initial value?

R04 · Vectors. Chasles' relation; available from S3. Simplify $\overrightarrow{AB} + \overrightarrow{BC} - \overrightarrow{AC}$.

R05 · Coordinates. Subtraction; available from S3. For A=(1;2), B=(4;-1), calculate $\overrightarrow{AB}$ and midpoint.

R06 · Collinearity. Proportionality; available from S3. Compare the vectors $(2;3)$ and $(-4;-6)$.

R07 · Python. For loop; available from S4. What values does k take in range(2,5) ?

R08 · Subscripts. Sums; available from S4. How many terms are in $\sum_{k=3}^{8} u_k$?

R09 · Mental calculation. Geometric sequence; available from S4. Calculate $1 + 2 + 4 + 8 + 16$. Give two methods.

R10 · Logic. Quantifiers; available from S5. Negate "all the terms are positive or zero".

R11 · Proof. Induction; available from S5. Which step is missing if we have only proved $P(n) \Rightarrow P(n+1)$?

R12 · Logic. Implication; available from S5. Is the converse of "x=2 implies $x^2 = 4$" true?

R13 · Vectors. Combination; available from S6. Expand 2 $(\vec{\imath} + \vec{k}) - 3(\vec{\jmath} + \vec{k})$.

R14 · Geometry. Basis; available from S6. Can a triple of coplanar vectors be a basis of space?

R15 · Error. Perspective; available from S6. Two lines cross on the drawing of a solid. Do the lines necessarily intersect?

R16 · Inequalities. Order; available from S7. From $1 \leq x < 4$, deduce bounds for $-2x + 3$.

R17 · Proof. Induction; available from S7. In $\sum_{k=0}^{n+1} k$, what term is added to the sum up to n?

R18 · Counter-example. Monotonicity; available from S7. Give an increasing sequence all of whose terms are negative.

R19 · Method. Affine sequence; available from S11. For $u_{n+1} = u_n/2 + 3$, which const. must be subtracted to get a geometric sequence?

R20 · Calculation. Fractions; available from S11. Simplify $\frac{1}{n+1} - \frac{1}{n+2}$ for $n \geq 0$.

R21 · Geometry. Plane; available from S12. Are two collinear directions enough to determine a plane?

R22 · Reading a band. Absolute value; available from S14. Translate $|u_n - 3| < 0{,}1$ as dbl. inequality.

R23 · Threshold. Inequality; available from S14. What is the first natural number n such that $1/(n+1) < 1/10$?

R24 · Method. Limits; available from S16. For $(3n^2+1)/(n^2+2n)$, which transformation to use?

R25 · Limit. Geometric sequence; available from S19. Compare the limits of 0.8^n & $(-0{,}8)^n$.

R26 · Counter-example. Convergence; available from S20. Does a bounded sequence always converge?

R27 · Theorem. Monotone convergence; available from S20. Must a rising sequence bounded above by 5 have limit 5?

R28 · Calculation. Roots; available from S21. Simplify $\sqrt{x^2}$ when x<0.

R29 · Sign. Quotient; available from S23. What is the sign of $1/(x-2)$ when $x < 2$? What limit at 2^- ?

R30 · Method. Conjugate; available from S24. Transform $\sqrt{x+1} - 1$ to study the quotient by x at 0.

R31 · Choice. Counting; available from S22. Three tops and two pairs of trousers: how many outfits?

R32 · Order. Lists; available from S25. How many three-letter codes using A,B,C with repetition?

R33 · No repetition. Lists; available from S25. Same question without repetition.

R34 · Method. Comparative growth; available from S26. Give the limit of x^2e^{-x} at $+\infty$.

R35 · Differentiation. Product; available from S29. Diff. $(x+1)e^x$.

R36 · Error. Composition; available from S29. Correct the proposed derivative $((2x+1)^3)' = 3(2x+1)^2$.

R37 · Graph. Tangent; available from S29. A tangent passes through (0;1) and (2;5). What is its slope?

R38 · Reading. Convexity; available from S29. If $f'' < 0$ on I, what can be said about f?

R39 · Counter-example. Inflection; available from S29. $f''(0) = 0$ enough for an inflection?

R40 · Continuity. Value at a point; available from S31. For $g(x) = (x^2 - 1)/(x - 1)$ off 1, what value should be given at 1 for continuity?

R41 · Theorem. IVT; available from S32. What does strict monotonicity add to the existence given by the IVT?

R42 · Python. Bisection; available from S32. After halving an interval of width 2 three times, what is its width?

R43 · Combinatorics. Subsets; available from S30. Count the teams of two among five.

R44 · Identity. Pascal; available from S35. Calculate $\binom{5}{2} + \binom{5}{3}$ without factorials.

R45 · Logic. Complement; available from S35. Why count "all minus none" for "at least one"?

R46 · Probabilities. Conditioning; available from S36. If P(A)=0.4 and $P_A(B) = 0,3$, calculate $P(A \cap B)$.

R47 · Probabilities. Reversal; available from S36. Are numbers $P_A(B)$ and $P_B(A)$ always equal?

R48 · Event. Integers; available from S36. Give the complement of "at least three successes".

R49 · Model. Binomial; available from S38. State the conditions of a Bernoulli scheme.

R50 · Probabilities. Binomial; available from S38. For X of law $\mathcal{B}(5; p)$, give P(X=0).

R51 · Mean. Binomial; available from S39. For X of law $\mathcal{B}(20; 0{,}3)$, give E(X) and V(X).

R52 · Bounds. Probability; available from S41. Express P(X>4) with a cumulative probability.

R53 · Bounds. Probability; available from S41. Express $P(2 \leq X \leq 5)$ as a difference of cumulatives.

R54 · Domain. Logarithm; available from S44. Determine the domain of ln $(3 - x)$.

R55 · Calculation. Logarithm; available from S44. Simplify ln 12 $-$ ln 3.

R56 · Error. Logarithm; available from S44. Test ln $(a + b) = \ln a + \ln b$ with a=b=1.

R57 · Differentiation. Logarithm; available from S47. Differentiate ln $(1 + x^2)$.

R58 · Limit. Logarithm; available from S49. Give the limit of $x \ln x$ at 0^+.

R59 · Inequality. Threshold; available from S50. In $q^n < c$ with 0<q<1 and 0<c<1, which direction after division by ln q?

R60 · Coordinates. Scalar product; available from S48. Calculate $(1;2;0)\cdot(3;-1;4)$ in orthonormal frame.

R61 · Angle. Scalar product; available from S48. A negative dot product indicates what type of angle between two non-zero vectors?

R62 · Normal. Plane; available from S51. How many non-collinear directions of the plane must be tested against a candidate normal?

R63 · Projection. Geometry; available from S54. State the two conditions for the foot of the perpendicular to a plane.

R64 · Reverse calculation. Antiderivatives; available from S52. Give an antiderivative of 3 e^{2x}.

R65 · Error. Antiderivatives; available from S53. Can two antiderivatives on an interval differ by a non-constant function?

R66 · Model. Differential equation; available from S55. Solve y'=2y with y(0)=3.

R67 · Units. Context; available from S55. Convert 1.5 h to minutes, then 0.1 min to seconds.

R68 · Python. Euler's method; available from S55. For y'=-y, y(0)=1 and h=0.1, calculate the first Euler step.

R69 · Sign. Integral; available from S59. Can a function that is negative on [a;b], a<b, have a positive integral?

R70 · Calculation. Integral; available from S61. Calculate $\int_0^1 2x\,\mathrm{d}x$.

R71 · Units. Integral; available from S63. Integrating a flow rate in L/min over minutes gives which unit?

R72 · Mean. Integral; available from S63. If $\int_2^5 f = 12$, what is the mean value on [2;5]?

R73 · Transformation. Variance; available from S62. If V(X)=5, give V(2X+3).

R74 · Mean. Independence; available from S64. For n independent copies with variance 9, give the variance of the mean.

R75 · Error. Spread; available from S64. Multiplying the size of an independent sample by 4 divides its standard deviation by what?

R76 · Parameter. Line; available from S65. For M(t)=(1+t;2-t;3t), give a point and a direction.

R77 · Normal. Plane; available from S67. Give a normal vector to the plane 2x-y+3z=5.

R78 · Membership. Plane; available from S67. Does the point (1;0;1) belong to 2x-y+3z=5?

R79 · Method. By parts; available from S66. In $\int x^2e^x\,\mathrm{d}x$, which factor should be differentiated to simplify?

R80 · Bounding. Rectangles; available from S66. For increasing f, do the left-hand rectangles give a lower bound or an upper bound?

R81 · Case. Intersection; available from S70. After substituting a line into a plane, what does 0=1 mean? And 0=0?

R82 · Error. Parameters; available from S70. Can the same parameter be used for two lines to find their intersection?

R83 · Concentration. Chebyshev; available from S73. With variance 4, which bound for $P(|X - E(X)| \geq 4)$?

R84 · Scope. LLN; available from S74. Does the law of large numbers guarantee getting closer at each new draw?

R85 · Angles. Première; available from S4. Convert 90 degrees to radians and give the coordinates of the associated point.

R86 · Circle. Signs; available from S6. On $[0; 2\pi]$, where is sin x positive or zero?

R87 · Period. Trigonometry; available from S76. Give the smallest positive period of sin(3x).

R88 · Differentiation. Composition; available from S76. Differentiate cos(2x).

R89 · Equation. Circle; available from S78. Solve cos x=1/2 on $[-\pi; \pi]$.

R90 · Antiderivative. Trigonometry; available from S79. Calculate $\int_0^{\pi/2} \sin x \, \mathrm{d}x$.

Suggested routines per session

Session	Questions	Follow-up
S2	R01, R02, R03	Answer without notes; explain a method; note the relevant follow-up.
S3	R04, R05, R06	Answer without notes; explain a method; note the relevant follow-up.
S4	R07, R08, R09	Answer without notes; explain a method; note the relevant follow-up.
S5	R10, R11, R12	Answer without notes; explain a method; note the relevant follow-up.
S6	R13, R14, R15	Answer without notes; explain a method; note the relevant follow-up.
S7	R16, R17, R18	Answer without notes; explain a method; note the relevant follow-up.
S8	R86, R85, R01	Answer without notes; explain a method; note the relevant follow-up.
S10	R02, R03, R04	Answer without notes; explain a method; note the relevant follow-up.
S11	R19, R20, R05	Answer without notes; explain a method; note the relevant follow-up.
S12	R21, R06, R07	Answer without notes; explain a method; note the relevant follow-up.
S13	R08, R09, R10	Answer without notes; explain a method; note the relevant follow-up.
S14	R22, R23, R11	Answer without notes; explain a method; note the relevant follow-up.
S15	R12, R13, R14	Answer without notes; explain a method; note the relevant follow-up.

Session	Questions	Follow-up
S16	R24, R15, R16	Answer without notes; explain a method; note the relevant follow-up.
S17	R17, R18, R86	Answer without notes; explain a method; note the relevant follow-up.
S19	R25, R85, R19	Answer without notes; explain a method; note the relevant follow-up.
S20	R26, R27, R20	Answer without notes; explain a method; note the relevant follow-up.
S21	R28, R21, R22	Answer without notes; explain a method; note the relevant follow-up.
S22	R31, R23, R24	Answer without notes; explain a method; note the relevant follow-up.
S23	R29, R25, R26	Answer without notes; explain a method; note the relevant follow-up.
S24	R30, R27, R28	Answer without notes; explain a method; note the relevant follow-up.
S25	R32, R33, R31	Answer without notes; explain a method; note the relevant follow-up.
S26	R34, R29, R30	Answer without notes; explain a method; note the relevant follow-up.
S27	R32, R33, R34	Answer without notes; explain a method; note the relevant follow-up.
S29	R35, R36, R37	Answer without notes; explain a method; note the relevant follow-up.
S30	R43, R38, R39	Answer without notes; explain a method; note the relevant follow-up.
S31	R40, R35, R36	Answer without notes; explain a method; note the relevant follow-up.
S32	R41, R42, R37	Answer without notes; explain a method; note the relevant follow-up.
S33	R43, R38, R39	Answer without notes; explain a method; note the relevant follow-up.
S35	R44, R45, R40	Answer without notes; explain a method; note the relevant follow-up.
S36	R46, R47, R48	Answer without notes; explain a method; note the relevant follow-up.
S37	R41, R42, R44	Answer without notes; explain a method; note the relevant follow-up.
S38	R49, R50, R45	Answer without notes; explain a method; note the relevant follow-up.
S39	R51, R46, R47	Answer without notes; explain a method; note the relevant follow-up.
S40	R48, R49, R50	Answer without notes; explain a method; note the relevant follow-up.
S41	R52, R53, R51	Answer without notes; explain a method; note the relevant follow-up.
S42	R52, R53, R01	Answer without notes; explain a method; note the relevant follow-up.
S43	R04, R02, R03	Answer without notes; explain a method; note the relevant follow-up.
S44	R54, R55, R56	Answer without notes; explain a method; note the relevant follow-up.
S47	R57, R54, R55	Answer without notes; explain a method; note the relevant follow-up.
S46	R56, R57, R05	Answer without notes; explain a method; note the relevant follow-up.

Session	Questions	Follow-up
S48	R60, R61, R07	Answer without notes; explain a method; note the relevant follow-up.
S49	R58, R60, R61	Answer without notes; explain a method; note the relevant follow-up.
S50	R59, R58, R06	Answer without notes; explain a method; note the relevant follow-up.
S51	R62, R59, R10	Answer without notes; explain a method; note the relevant follow-up.
S52	R64, R62, R08	Answer without notes; explain a method; note the relevant follow-up.
S53	R65, R64, R09	Answer without notes; explain a method; note the relevant follow-up.
S54	R63, R65, R11	Answer without notes; explain a method; note the relevant follow-up.
S55	R66, R67, R68	Answer without notes; explain a method; note the relevant follow-up.
S56	R63, R66, R67	Answer without notes; explain a method; note the relevant follow-up.
S60	R68, R13, R14	Answer without notes; explain a method; note the relevant follow-up.
S58	R12, R16, R15	Answer without notes; explain a method; note the relevant follow-up.
S59	R69, R17, R18	Answer without notes; explain a method; note the relevant follow-up.
S61	R70, R69, R86	Answer without notes; explain a method; note the relevant follow-up.
S62	R73, R70, R19	Answer without notes; explain a method; note the relevant follow-up.
S63	R71, R72, R73	Answer without notes; explain a method; note the relevant follow-up.
S64	R74, R75, R71	Answer without notes; explain a method; note the relevant follow-up.
S65	R76, R72, R74	Answer without notes; explain a method; note the relevant follow-up.
S66	R79, R80, R75	Answer without notes; explain a method; note the relevant follow-up.
S67	R77, R78, R76	Answer without notes; explain a method; note the relevant follow-up.
S70	R81, R82, R79	Answer without notes; explain a method; note the relevant follow-up.
S69	R80, R77, R78	Answer without notes; explain a method; note the relevant follow-up.
S72	R81, R82, R85	Answer without notes; explain a method; note the relevant follow-up.
S73	R83, R20, R22	Answer without notes; explain a method; note the relevant follow-up.
S74	R84, R83, R21	Answer without notes; explain a method; note the relevant follow-up.
S75	R84, R24, R23	Answer without notes; explain a method; note the relevant follow-up.
S76	R87, R88, R26	Answer without notes; explain a method; note the relevant follow-up.
S77	R87, R88, R25	Answer without notes; explain a method; note the relevant follow-up.
S78	R89, R28, R27	Answer without notes; explain a method; note the relevant follow-up.

Session	Questions	Follow-up
S79	R90, R89, R31	Answer without notes; explain a method; note the relevant follow-up.
S80	R90, R30, R29	Answer without notes; explain a method; note the relevant follow-up.
S81	R34, R32, R33	Answer without notes; explain a method; note the relevant follow-up.
S83	R35, R36, R37	Answer without notes; explain a method; note the relevant follow-up.
S85	R43, R38, R39	Answer without notes; explain a method; note the relevant follow-up.
S86	R40, R44, R41	Answer without notes; explain a method; note the relevant follow-up.
S88	R42, R45, R46	Answer without notes; explain a method; note the relevant follow-up.
S89	R47, R49, R50	Answer without notes; explain a method; note the relevant follow-up.
S91	R48, R51, R52	Answer without notes; explain a method; note the relevant follow-up.
S92	R53, R57, R56	Answer without notes; explain a method; note the relevant follow-up.

Fluency solutions

R01. 10 and 13. The index replaces n.

R02. 1/8 and 4, by the laws of exponents.

R03. No: 0.8 $\times 1{,}2 = 0.96$, i.e. an overall decrease of 4%.

R04. The zero vector, by Chasles.

R05. $(3;-3)$; midpoint $(5/2;1/2)$.

R06. The second is -2 times the first: they are collinear, in opposite directions.

R07. 2,3,4. The final endpoint is excluded.

R08. Six: $8 - 3 + 1$.

R09. 31 ; direct addition or $2^5 - 1$.

R10. There is a rank n such that $u_n < 0$.

R11. The base case at a specified starting index.

R12. No: x=-2 satisfies $x^2 = 4$ without being 2.

R13. $2\vec{\imath} - 3\vec{\jmath} - \vec{k}$.

R14. No: its combinations stay in a vector plane.

R15. No: projection can create an apparent intersection.

R16. $-5 < -2x + 3 \leq 1$; multiplying by -2 reverses the inequalities.

R17. n+1, not n.

R18. $u_n = -1/(n+1)$.

R19. 6, since $6 = 6/2 + 3$.

R20. $1/((n+1)(n+2))$, strictly positive.

R21. No; you need two non-collinear directions and a point.

R22. $2{,}9 < u_n < 3{,}1$.

R23. 10 ; at index 9, the gap is exactly 1/10.

R24. Divide top and bottom by n^2 ; the limit is 3.

R25. Both tend to 0 because the absolute value of the common ratio is less than 1.

R26. No: $(-1)^n$ is bounded but its terms alternate between -1 and 1.

R27. No; it converges to a limit at most equal to 5, for example a constant sequence equal to 2.

R28. $-x$, since $\sqrt{x^2} = |x|$.

R29. Negative; limit $-\infty$.

R30. $\sqrt{x+1} - 1 = x/(\sqrt{x+1}+1)$; the quotient tends to 1/2.

R31. Six, by the multiplicative principle.

R32. $3^3 = 27$; the positions are distinguished.

R33. $3! = 6$; three choices, then two, then one.

R34. 0, by comparing growth rates; the form $\infty \times 0$ alone is not conclusive.

R35. $(x+2)e^x$.

R36. Factor 2 is missing: derivative 6 $(2x+1)^2$.

R37. 2 ; vertical change 4 divided by horizontal change 2.

R38. f is concave on I; this does not determine the sign of f'.

R39. No: $f(x) = x^4$ is convex on both sides of 0.

R40. 2, since g(x)=x+1 away from 1.

R41. Uniqueness on the interval considered.

R42. $2/8 = 1/4$.

R43. $\binom{5}{2} = 10$; each ordered pair would have been counted twice.

R44. $\binom{6}{3} = 20$, by Pascal's rule.

R45. The complement of "at least one" is "none"; the cases are disjoint and exhaustive.

R46. 0.12, product along the branch.

R47. No; reference populations differ.

R48. At most two successes.

R49. Fixed number of trials, independence, two outcomes, same probability of success.

R50. $(1-p)^5$.

R51. 6 and 4.2.

R52. $1-P(X\leq 4)$ for integer X.

R53. $P(X\leq 5)-P(X\leq 1)$.

R54. $]-\infty;3[$.

R55. $\ln 4 = 2\ln 2$.

R56. The left-hand side equals ln 2, the right-hand side 0; the identity is false.

R57. $2x/(1+x^2)$ on $\mathbb{R}$.

R58. 0 ; set t=-ln x to obtain $-te^{-t}$.

R59. $n > \ln c/\ln q$; the direction is reversed because ln q<0.

R60. 1.

R61. An obtuse or straight angle, in $]\pi/2;\pi]$.

R62. Two; also check that the candidate is nonzero.

R63. Membership of the plane, and the direction joining the point to the foot is orthogonal to the plane.

R64. $\frac{3}{2}e^{2x}$; differentiation provides the check.

R65. No: the derivative of their difference is zero on the interval.

R66. $y(t)=3e^{2t}$.

R67. 90 min and 6 s.

R68. $1-0{,}1\times 1 = 0{,}9$ at t=0,1.

R69. No; the positivity of the integral applied to -f gives $\int_a^b f \leq 0$.

R70. 1, by the antiderivative x^2.

R71. Litres.

R72. 4 ; divide by the length 3.

R73. 20; the factor is 2^2, a translation does not change the variance.

R74. $9/n$.

R75. 2, since the standard deviation is proportional to $1/\sqrt{n}$.

R76. (1;2;0) and (1;-1;3).

R77. (2;-1;3), or any non-zero multiple.

R78. Yes, because 2+0+3=5.

R79. The polynomial x^2 ; its degree decreases.

R80. A lower bound, since each rectangle stays below the curve.

R81. No intersection; then the line lies entirely in the plane.

R82. No, each line is traversed independently.

R83. $1/4$.

R84. No; it concerns the probability of a fixed deviation as the sample size increases.

R85. $\pi/2$; (0;1).

R86. On $[0;\pi]$ and at point 2 π.

R87. $2\pi/3$.

R88. $-2\sin(2x)$.

R89. $x = -\pi/3$ or $x = \pi/3$.

R90. 1, by the antiderivative -cos x.